LESSON PLAN

PROPELLER THEORY (7)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various checks on propeller on the ground during ground run
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various checks on propeller on the ground during ground run
5. Specific Objectives	7.1.2.1 Trainee will be able to explain propellers and its associated maintenance practice

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Conduct ground run checks on propeller; Safety: Ensure clear area and proper RPM limits; Supervision: Instructor monitors all readings and safety protocols
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of a ground run check?
(b) List three safety precautions before starting a ground run.
(c) Which instruments are typically monitored during a propeller ground run?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Pre-run safety inspections • Engine start and idle verification • RPM gauge reading and stability • Vibration and noise assessment • Blade tracking visual check	L & Ds	PPT, IB	15 min
(b)	• Thrust bearing temperature monitoring • Oil pressure and temperature checks • Fuel flow and mixture settings • Electrical system load testing • Cooling system efficiency verification	L & Ds	PPT, PDF	20 min
(c)	• Post-run shutdown procedures • Final visual inspection for damage • Documentation of run data • Comparison with standard limits • Reporting anomalies to maintenance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the significance of monitoring thrust bearing temperature during a ground run.
(b) Describe the correct procedure for documenting ground run data for maintenance records.

12. Home Assignment and Follow Up	Prepare a brief report outlining the key differences between ground run checks and in-flight propeller inspections, referencing LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	General precautions to be observed while working on propeller, ground run	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min